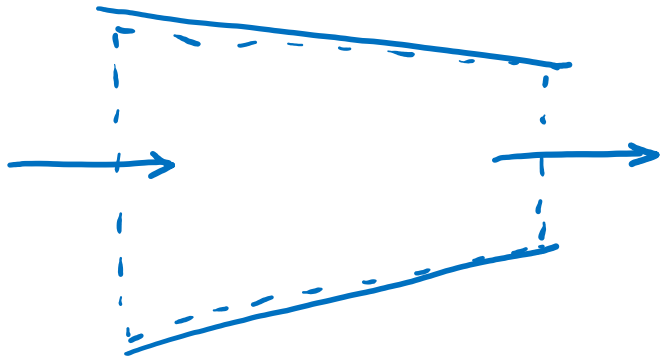




ME 300:
Engineering Thermodynamics

Conservation Principles

Conservation basics



→ Energy
→ Mass

} → finite time
→ instantaneous

Finite time formulation

→ conservation over a finite time interval: $\Delta t = \underline{t_2} - \underline{t_1}$

→ define system: $\Delta X_{\text{sys}} = X(t_2) - X(t_1)$
 $= X_{\text{in}} - X_{\text{out}} + X_{\text{gen}}$

$\uparrow$ $\uparrow$ $\uparrow$ $\uparrow$
change in in out generated
conserved inside system
quantity

$$\Delta E_{\text{sys}} = E_{\text{in}} - E_{\text{out}} + \underline{\underline{E_{\text{gen}}}}$$

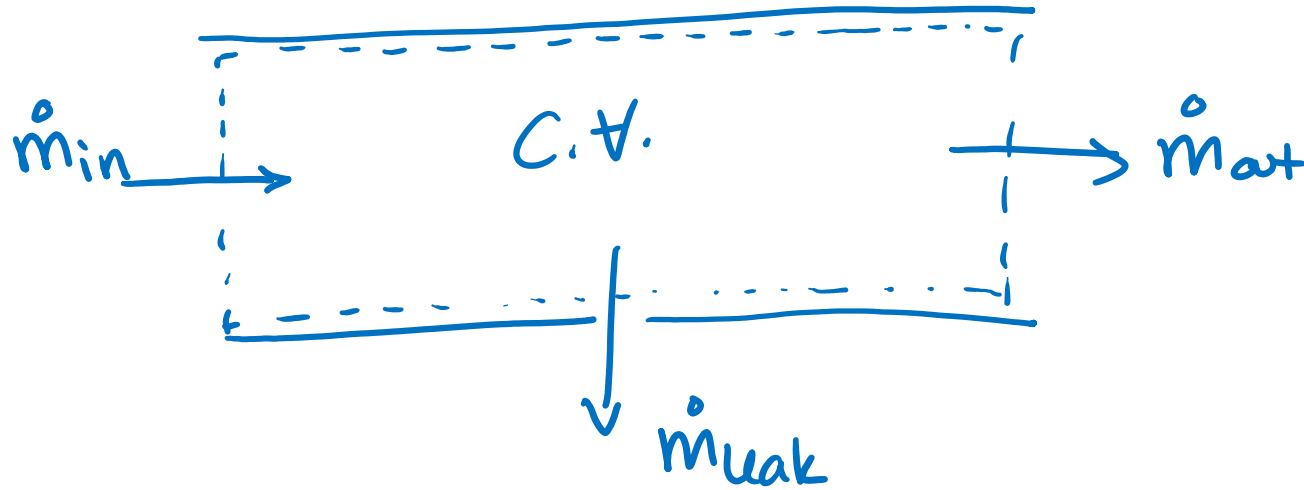
x

Instantaneous formulation $\rightarrow$ rates $\Rightarrow$ flow
 $\Rightarrow$ leakage
 $\Rightarrow$ transfer

System X_{sys}

$$\frac{dX_{\text{sys}}}{dt} = \dot{X}_{\text{in}} - \dot{X}_{\text{out}} + \dot{X}_{\text{gen}} \quad \sim \left[\frac{\text{units}}{\text{sec}} \right]$$

Example: leaky pipe



$$\frac{dm_c}{dt} = \underbrace{\dot{m}_{in}}_{\text{rate added}} - \underbrace{\dot{m}_{out} + \dot{m}_{leak}}_{\text{rate subtracted.}}$$